BVM ENGINEERING COLLEGE [AN AUTONOMOUS INSTITUTION]

201CP: OBJECT ORIENTED PROGRAMMING WITH C++

CREDITS - 4 (L_c , T_c , P_c : 3, 0, 1)

2nd Year, B. Tech. (Computer Engineering)

Course Objective: To impart the knowledge about principles of Object Oriented Programming paradigm using C++.

Teaching and Assessment Scheme:

Teaching Scheme (Hours per Week)			Credits	Assessment Scheme				Total Marks
				Theory		Practical		
L	T	P	C	ESE	CE	ESE	CE	
3	0	2	4	60	40	20	30	150

Course Contents:

Unit No.	Topics	Teaching Hours
1	**Basics:** Introduction to Object Oriented Programming (OOP); Procedural Vs. OOP; Principles of OOP; Benefits and Applications of OOP; Program Structure; Namespace; Identifiers; Variables; Constants; Operators; Typecasting; Control Structures	06
2	**Objects and Classes:** Basics of Object and Class; Private and Public members; Static Data Member and Member Function; Constructors and their types; Destructors; Type Conversion; Pointers and Objects; this pointer; New and Delete operators. Arrays of Objects; Reference variables	06
3	**Functions & Overloading:** Simple functions; Call and Return by Reference; Inline functions; Macro Vs. Inline functions; Overloading of functions; Copy constructor; Move constructor; Operator overloading; Default arguments; constant arguments and constant function; Friend functions.	07
4	**Inheritance and Dynamic Polymorphism:** Concept of Inheritance; Types of Inheritance: Single, Multiple, Multilevel, Hierarchical, Hybrid; Protected members; Method Overriding; Virtual Base Class; Virtual and Pure Virtual functions; Implementing Dynamic Polymorphism	08
5	**Exception Handling:** Review of Traditional Error Handling; Basics of Exception Handling; Exception Handling Mechanism; Throwing and Catching Mechanisms; Rethrowing an Exception; Specifying Exceptions	04
6	**I/O and File Management:** Concept of Streams; ‘cin’ and ‘cout’; Overloading of Inserter and Extractor operators; C++ Stream Classes; Unformatted and Formatted I/O; Manipulators; File stream and C++ classes; File Management functions; File modes; Binary and Random Files	06

Unit No.	Topics	Teaching Hours
7	**Templates, STL & RTTI:** Introduction; Need and Use of Templates; Function templates and Class templates; STL: Introduction, Containers, Algorithms and Iterators; RTTI: Introduction, user defined namespaces	05
	Total	42

Some of the above or related topics may also be covered through – Self Learning or Co-Curricular activities (like Workshops / Peer Learning / Professor of Practice / Expert talks)

List of References:

1. E Balagurusamy, “Object Oriented Programming with C++”, TMH (E-book available on the BVM Intranet)
2. Herbert Schlitz, “The Compete Reference C++”, TMH
3. Deitel, “C++: How to Program”, PHI
4. Scott Meyers, “Effective C++”, Addison-Wesley
5. Debasish Jana, “C++ and Object Oriented Programming Paradigm”, PHI
6. Ashok Kamthane, “Object Oriented Programming with ANSI and Turbo C++”, Pearson

Course Outcomes (COs):

After successful completion of the course; the student will be able to:

1. Differentiate between Object Oriented Programming and Procedural Programming Paradigms
2. Demonstrate features of Object Oriented Programming like Encapsulation, Inheritance, Polymorphism etc by implementing them using C++
3. Design a solution of a given problem using Object Oriented Programming concepts
4. Implement the solution of a given problem using C++
5. Implement features like Exception handling, Templates, STL and file management in the solution
6. Develop and improve Logical Reasoning and Programming skills